Evaluation of auditory spatial impression in performance spaces

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The three-dimensional time-varying output of a computational model of the human binaural system was investigated with the aim of extracting perceptual patterns which correlate with auditory spatial impression. To this end, the same stimuli ─ generated in a precisely defined way ─ with which the model was excited were also presented to human listeners which evaluated the auditory source width (ASW), i.e. the horizontal extent of the auditory event.

Subjects were trained beforehand for these experiments as the discrimination between source- and room-related parameters of auditory events is not a common task to them. Stimuli were varied in temporal resolution to evaluate and fit the temporal resolution of the subjects' binaural system.

Since ASW has been identified in the field to be a relevant perceptual attribute of spaces for musical performances, applying a binaural model for its estimation is a useful tool for the prediction of the auditory quality of these spaces. In fact, temporal fluctuations in the binaural-activity maps rendered by the model seem to be a valuable indicator of the spatial extent of auditory events.

1 Introduction

A main criterion being essential for an auditory preference of a concert hall by listeners is a good spatial perception within the audience space. One of the prominent problems in this field of room acoustics is to provide clear guidelines and algorithms on how one can evaluate the spatial perception of an environment in a fairly universal way by an instrumental measurement and an instrumental evaluation, i.e. as independent as possible from an individual listener. Established binaural measures like interaural cross-correlation (*IACC*) or lateral energy fraction (*LEF*), recommended in ISO3382 [12], show a significant dependency on the measurement position as investigated by de Vries et al. in 2001 [26]. Their documentation of measurements in two concert halls demonstrated that a measurement distance of less than one meter within the center of a concert hall can cause differences greater than 0.9 in *IACC* results, whose range is from zero to one.

The results of monaural parameters, which, e.g., take into account the balance between early and late arriving energy, like clarity (C_{80}), definition (D_{50}), early decay time (*EDT*) or center time (T_S), often fail in correlating well with perceptual attributes, see Hess et al., 2003 [11]. New parameters, which combine aspects of directional-dependent and directional-independent parameters, investigated by Soulodre et al. in 2003 [24], motivated by insufficiencies mentioned above, however, did not show fully satisfying results either. The most common measure reverberation time (T_{60}) is also unable to give a clear answer on whether a room sounds good or not. A concert hall with a T_{60} of two seconds may sound good, but it does not necessarily have to sound good.

Taking this situation into account, there still remains some doubt on whether the ranking of concert halls based on such measures as listed above ─ although certainly correct from a signal-processing point of view ─ see, e.g., Beranek, 2003 [1] ─ really reflects the perceptual rank order of the auditory preference of the compared halls. The underlying problem is that these parameters basically describe a room's physical spatial attributes, but what really matters is the human perception of spatial sound. The aim of this work is to investigate this perception by introducing a method as initiated by Lindemann in 1982 [17] and Bilsen in 1994 [2], which is based on simulating the complete pathway of human auditory signal processing.

Spatial perception of humans in reverberant rooms can be subsumed by acoustical properties which are attributed (i) to the sound source, and (ii) to the environmental context. In a characteristic way auditory events are perceived as spread out in contrast to the visual extent of the sources radiating sound. Immersion or envelopment by auditory events, which is linked to the latter, can be perceived. This perception is caused by interaural parameters and therefore only possible using binaural hearing.

General agreement exists that reflections of direct sound arriving at the listeners ears are causing both. Depending on the size of the room and materials used for building the room, the location of the sound source will be perceived as slightly ambiguous and broader regarding its size. It is well accepted that this is caused by early lateral reflections, whereas "early" stands for a sound-arrival within 1 to 80ms after direct sound arrival in a concert hall. Together with the direct sound the reflections are summed up and seem to come from the direction where the sound source is located. It is also generally understood that a temporal and spatial separation of energy arriving late after direct sound is perceived as coming from around a listener, causing a perception of being enveloped by the sound. "Late" in this context means an arrival-time of sound later than 80 ms after direct sound arrival.

In 1968 de Villiers Keet [25] pointed out that "broadening of the source" is a very important criterion in concert hall acoustics, whose presence or absence can cause the difference between a good and an excellent sounding hall. It was barely 1960 when Lochner and de Villiers Keet [19] differentiated between perception of sound in a concert hall which is separable in direction of incidence, and a "sensation of depth and presence" that is created "on account of delayed reflections or echoes". The former they depicted as "apparent positions of sound sources", the latter as "ambience". Based on the work of Kuhl, 1978, [16] and work of Reichardt and Lehmann in 1978 [23], Blauert and Lindemann in 1986 [5] introduced the terms "auditory spatial impression" and "reverberance". Auditory spatial impression was denominated as "characteristic spatial spreading of the auditory events that is predominantly caused by early lateral reflections", whereas a larger amount of space is filled by auditory events than defined by the visual contours of the sound sources. Reverberance was termed as "a characteristic temporal slurring of the auditory events which results from late reflections and reverberation", causing a perception of the listener being enveloped by auditory events. Generally spoken, spatial impression is the "concept of the type and size of an actual or simulated space at which a listener arrives spontaneously, when he or she is exposed to an appropriate sound field". As first authors they pointed out the multidimensionality of these perceptual attributes. The ambiguousness of the terms "spaciousness", "spatial impression", "Räumlichkeit", "Schwimmbadeffekt" etc. was resolved when — returning to de Villiers Keet, 1968 [25] — Morimoto and Maekawa in 1989 [21] and 1990 [22] introduced the nowadays widely accepted terms "auditory source width" (ASW), denoting perception of the spatial extent of the auditory event, and "listener envelopment" (LEV), denoting the perception of the amount of envelopment by the auditory event(s).

In 1995 Bradley and Soulodre [6] experimentally confirmed this concept in modifying late parts of room-impulse responses in level and temporal structure, and presented them to subjects. They were able to show a clear separation between the ASW, which was formerly demonstrated to be related to the relative level of early lateral reflections, and LEV, which they demonstrated is related to level, arrival-direction, and distribution in time of late reflections, arriving later than 80 ms after direct sound arrival.

To come up to the multidimensionality of human spatial perception a visualization of the four dimensions of the outcome of a binaural model, the so-called binaural activity map, seems to be a promising approach. Lateralization, the amount of binaural activity, time and auditory frequency-bands are shown there. In 1979/82 Blauert [3] hypothesized that the delay of interaural phase considered as a function over frequency causes auditory spatial impression. Lindemann in 1982 [17] showed in a movie where he visualized binaural activity as a dynamic process, that an increased broadening of the auditory event is visualized by fluctuations in lateral deviation over time and frequency. In 1986 Blauert made his predication more general in presuming temporal and spatial incoherence as the main reason for spatial impression. Also Griesinger 1997 [9] supposed that temporal fluctuations are the most important criterion for spatial perception. Informal listening tests, however, showed that the fluctuation frequency depicted by the binaural model can not be perceived as fluctuating, i.e. a temporal smoothing or smearing is done by the human auditory system. This is referred to as binaural sluggishness. Modeling the binaural sluggishness by applying the model would deliver a measure for the spatial extent of the auditory event, the ASW.

2 The model algorithm

The binaural model algorithm applied in this work is signal driven, i.e., it has a strict bottom-up architecture. The algorithm is based on the well known physiologically-motivated algorithm to estimate interaural time differences (ITDs), as originally proposed by Jeffress in 1948 [15]. Lindemann [18] introduced contralateral inhibition elements to enhance this algorithm to sharpen the peaks in the binaural activity pattern and to suppress side-lobes which occur because of the periodic nature of the cross-correlation algorithm for band-pass signals. As a result other positions than those where the coincidence occurred are suppressed. A second objective introducing contralateral inhibition was to combine the influences of the ITDs and the interaural level differences (ILDs). The latter are of particular importance in the frequency range above 1.5 kHz. Thus the results of the cross-correlation algorithm become dependent on the ILDs. For a general overview of the cross-correlation and inhibition structure see Fig. 2.1. Note that monaural activity can be inhibited by binaural one, but not vice versa. Gaik in 1993 [7] introduced an amendment to simulate the localization in the horizontal plane in a more exact way. His extension allows to optimally process the natural combinations of ITDs and ILDs, as they are found in head-related transfer functions. For a more detailed description applying this algorithm to room-acoustical problems see Hess et al., 2003 [10]. A comprehensive overview can be found in Blauert, 1996 [4].

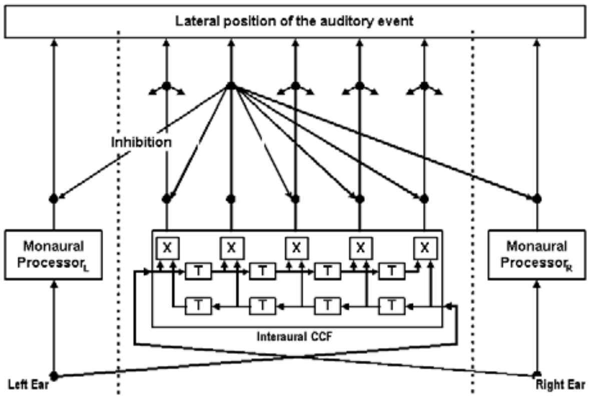

Figure 2.1: Cross-correlation function and inhibition structure of the binaural model, from [18].

3 Training

Since discrimination in source and room related parameters of auditory objects is not a common task, the subjects were trained beforehand in small groups of four persons in four training sessions on four consecutive days each, see Merimaa and Hess, 2004 [20]. Altogether 16 subjects, chosen based on an exploration by questionnaires and an audiometric screening, participated in the training. They listened to natural samples and noise samples over headphones. After hearing the samples the listeners had to draw the auditory events and to discuss their individual spatial perception. The training was double blinded in the sense that instructors did not know the order of the stimuli and wore no headphones. Also the moderation was done in a way to aid the subjects in expressing their spatial impression, however it was avoided to bias them in any way. The composition of the groups changed every session to guarantee that not a single dominating group member impacts the other ones. One assumption of [20] was that trained listeners seem to produce consistent results by drawing auditory objects. Therefore the horizontal spatial extent had to be marked in a compass wind-rose by one or two lines, depending on the number of auditory events the listeners perceived. The subjects had to learn both to get a clear idea of distinguishing between source-related and environmental-related auditory events and to localize them inside and outside the head. A mapping of the spatial extent of auditory events inside the head was necessary, as well as mapping of envelopment inside the head.

4 Experiment

In the experiment six trained listeners, three female and three male, aged between 20 and 25, served as paid subjects. They had to judge the spatial extent and the movement of the auditory events of interaural temporally varying frequency-modulated (Fm) broadband stimuli. The results had to be drawn into a custom built graphical user interface by mouse-operation. Only the pure influence of ITD-variation was intended to be judged by subjects. Therefore signals were directly presented via headphones avoiding room-acoustical side effects and an influence by head-related transfer functions (HRTFs).

4.1 Stimuli

Already in 1978 Grantham and Wightman [8] conducted a detection experiment, where they asked for perception of movement of an interaural temporally varying frequency-modulated stimulus in a two-interval forced-choice test. They observed that subjects detected the fluctuation up to a distinct threshold based on movement of the stimuli and above this threshold by the “width of the image”. Wightman and Kistler, 1992 [27], suggested from results of their evaluation, where the influences of ITDs were compared to those of ILDs, that low-frequency ITDs are dominant compared to other localization factors.

The stimuli were generated digitally by the algorithm shown in equations 4.1 and 4.2. Originally excogitated by [8], this method calculates two sinusoidally frequency-modulated noises with a certain maximum interaural time difference.

$$l(t) = \sum_{i=1}^{N} A \cdot \sin\left(\frac{2\pi \cdot f_i \cdot t}{Fs} + \theta_i + \pi \cdot f_i \cdot \Delta t \cdot \sin\left(\frac{2\pi \cdot Fm \cdot t}{Fs}\right)\right) \quad (4.1)$$

$$r(t) = \sum_{i=1}^{N} A \cdot \sin\left(\frac{2\pi \cdot f_i \cdot t}{Fs} + \theta_i - \pi \cdot f_i \cdot \Delta t \cdot \sin\left(\frac{2\pi \cdot Fm \cdot t}{Fs}\right)\right) \quad (4.2)$$

The leading and lagging part is continuously changed as a sinusoidal function of time. When presented directly to both ears, a varying interaural temporal difference with an opposite phase is caused. The bandwidth of the stimuli was 0.02 to 3.5 kHz.

4.2 Setup and Procedure

The experimental setup was arranged in a sound-attenuating booth. Stimuli were presented via STAX Lambda Pro headphones driven by a diffuse-field equalized headphone driver unit. The stimuli were stored on a PC and represented via an ASIO sound adaptor RME HDSP Multiface.

After explaining to the subjects how to operate the graphical user interface and to start an experimental run, instructions for the listening test were given to them in written form in German. In marking the auditory source width, i.e. the horizontal extent of the auditory event, they had to draw their individual spatial perception on a graphical user interface. They were instructed to mark the limits of the auditory event(s) by the end of line(s). If they perceived a movement of the stimulus they were instructed to draw two lines, where the outer limits designated the area within the stimuli were perceived. In this way a discrimination between moving and stationary auditory events was done. Since stimuli were represented without HRTFs, locatedness was intracranial. To give consideration to that fact, subjects had to draw inside the symbol of a head.

Each experimental pass consisted of twelve stimuli, which were each randomly assigned to a diagram in the user interface containing a compass wind-rose and the symbol of the head of the subject. Six runs of the experiment were conducted. The first run was considered practice and left out of the analysis.

Stimuli consisted of two parts: A dichotic signal of duration 1000 ms, followed by a pause or silence time of 1000 ms. These parts were continuously repeated until a “stop”-button was pressed. This procedure ensured that a full control of the temporal rules of repeated listening to the stimuli took place. The ITD in the beginning of each signal part was zero. As maximum of the fluctuating ITD Δt = 60 µs was chosen.

4.3 Results

The analysis was done by calculating means (limits of vertical rectangles) and upper and lower quartiles of the horizontal spatial extent, please refer to Fig. 4.1.

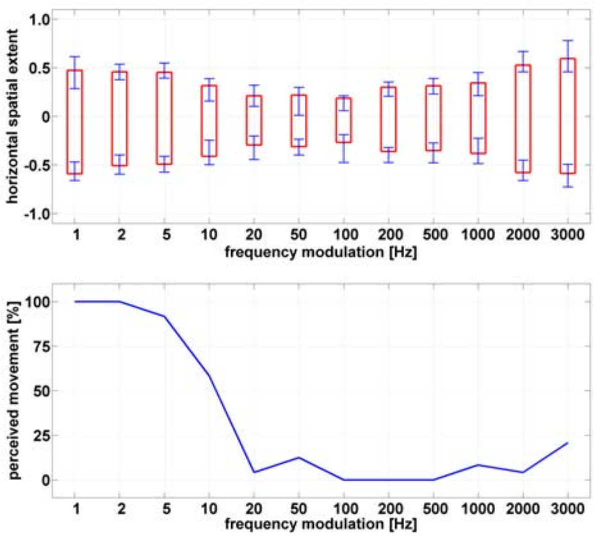

Figure 4.1: Experimental results of maximum interaural time difference Δt = 60 µs.

In order to transform the data from all listeners to the same scale, means and standard deviations of the judgments of each subject during each run were scaled to the mean and the standard deviation which were calculated over all subjects and all runs, as recommended by ITU [13] and [14]. The ASW is shown in the upper diagram, whereas the lower illustrates the absolute value of movement perception. The abscissa shows the modulation frequency of the stimuli and is equivalent in both diagrams. ASW-data was interpreted in the following way: Since the auditory source width consisted of a moving auditory event with left and right outer limit, the analysis of the extent was calculated from the left outer limit of the left boundary and the right outer limit of the right one. Within this area subjects perceived the auditory events in every case.

The movement analysis diagram shows a clear lowpass function. If a line is fitted to the threshold where movement perception changes from moving to stationary, a Fm of approximately 10 Hz can be identified where the line intersects the 50% movement line. A relatively symmetrical spatial extent was perceived, as can be extracted from the lateral deviation of the results of horizontal spatial extent. The ASW changes with modulation frequency. At low modulation frequencies a moving stimulus with a wide lateral deviation is perceived. Approaching the threshold where movement perception changes to a stationary signal perception, the width of the auditory event(s) gets more and more compact. Approximately at the threshold the signal is perceived as most compact. Up to a higher frequency modulation, the horizontal broadening grows again and is perceived as broadest at 3000 Hz modulation frequency.

5 Adaptation of the model

A simulation of the stimuli with the binaural model

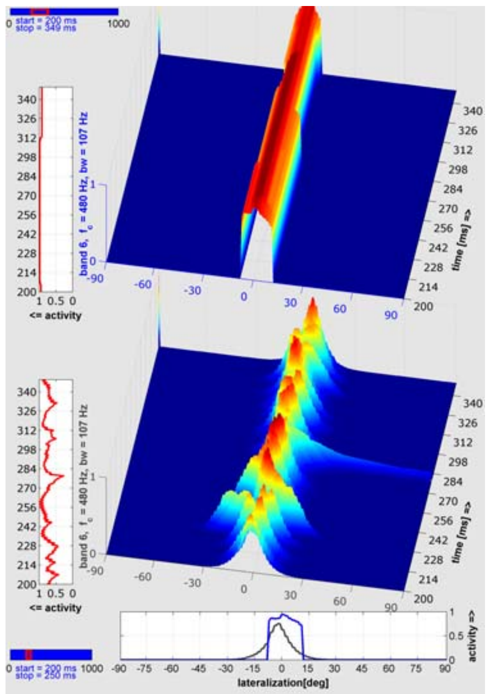

Figure 5.1: Model simulation of maximum ITD Δt = 60 µs, Fm = 100 Hz.

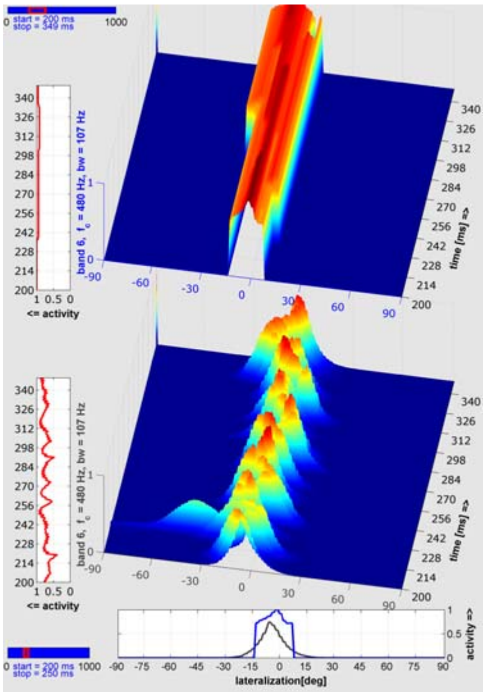

Figure 5.2: Simulation of Δt = 60 µs, Fm = 500 Hz.

shows a high fluctuation in neural activity, as depicted in the lower 3-D plot, the so-called binaural activity map (BAM). In figures 5.1 to 5.5 one auditory band, f_C = 480 Hz, bandwidth of 107 Hz, is visualized in the lower BAM directly as calculated by the model algorithm. Here the x-axis represents the lateral deviation from –90° to +90°, the y-axis the amount of neural activity, and the z-axis the running time in milliseconds. The upper BAM shows the same auditory

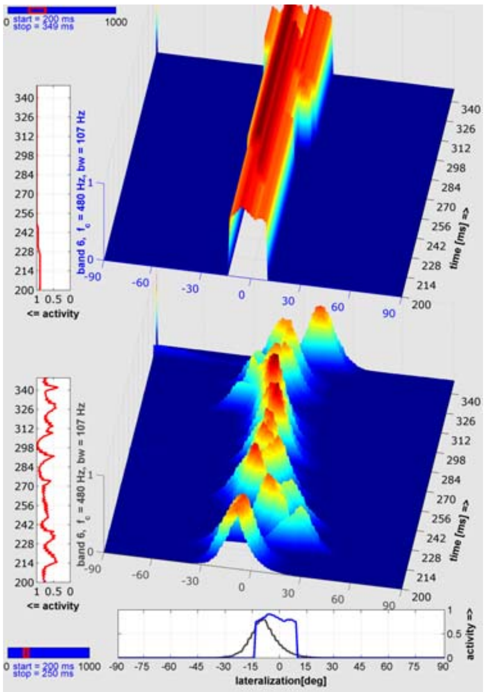

Figure 5.3: Simulation of Δt = 60 µs, Fm = 1000 Hz.

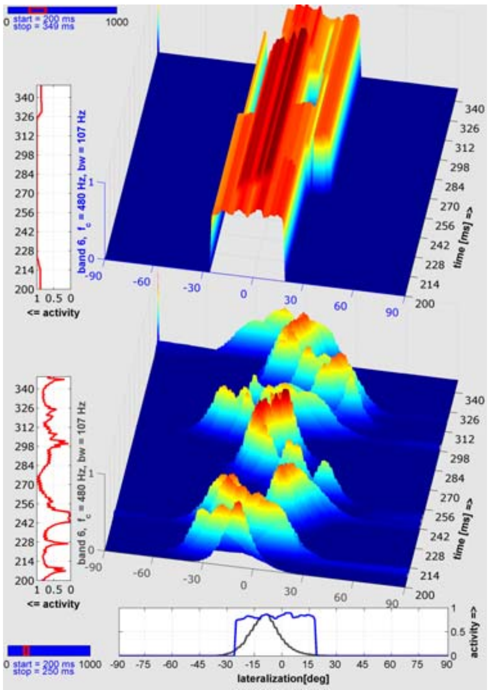

Figure 5.4: Simulation of Δt = 60 µs, Fm = 2000 Hz.

band, however, it was calculated in fitting the model algorithm to the experimental results: A temporal smoothing is done with a floating rectangular time window of 100 ms duration, according to a Fm of 10 Hz. As can be seen in the resulting lateral deviation depicting the spatial extent of the auditory event, an increase from Fm = 100 Hz to Fm = 3000 Hz, very similar to the experimental results, is identifiable. The limits of the lateral deviation, depicted also in the

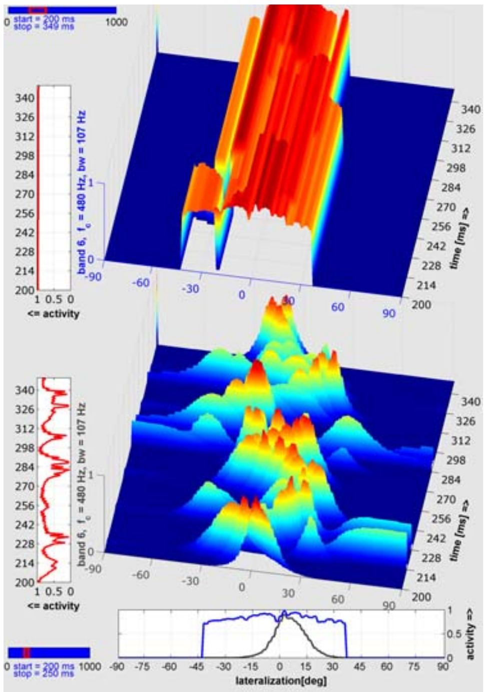

Figure 5.5: Simulation of Δt = 60 µs, Fm = 3000 Hz.

bottom plot of each figure, give a clear indication of the spatial extent of the auditory event in good correlation to the experimental results. Fm = 500 and 1000 were also in the experiment perceived as similar in spatial extent, whereas Fm = 2000 and 3000 strongly increased in spatial extent as indicated in the smoothed upper binaural activity map.

6 Conclusion

Experiments were conducted to characterize the horizontal spatial extent of artificially generated sinusoidally ITD-fluctuating broadband noises. As the maximum interaural time difference Δt = 60 µs was chosen. The results show that the perceived spatial extent depended on the modulation frequency, and followed a bandstop-function. The auditory events were perceived as widest in extent at the upper and lower end of the range, in which the modulation frequencies were tested. In the lower end movement was perceptible, and the subjects marked the area within they perceived the auditory objects moving. At the upper end most subjects perceived one fused auditory event which increased in extent proportional to the modulation frequency Fm. The threshold of the transition from movement to a stationary stimulus, however, characterized this modulation frequency where the signal was perceived by the subjects as most compact.

Applying this threshold to a floating time window for smoothing the 3-dimensional outcome ─ the so-called binaural activity map ─ of a localization model of the human auditory system, delivers reasonable results to visualize and to quantify the horizontal spatial extent of auditory events. This is nowadays called in literature auditory source width (ASW).

In the future the auditory source width of a performance space could be evaluated in convolving the above described stimuli signals with the binaural room-impulse responses of the halls. In simulating the binaural activity map and smoothing it like described above as a result the ASW is calculated. Its hoped that our algorithm will prove useful as a tool for room-acoustics analysis.

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